ML Classification Project Report

TITLE:

Breast Cancer Classification Using Machine Learning

1. Introduction

Machine Learning classification is a supervised learning technique used to predict categorical outcomes based on input features. In this project, a classification model was developed to predict whether a breast tumor is malignant (cancerous) or benign (non-cancerous).

The objective of this project is to compare the performance of two machine learning algorithms, Logistic Regression and Random Forest Classifier, and determine the most effective model for classification.

2. Objective

The main objectives of this project are:

- To perform data preprocessing and analysis.
- To split the dataset into training and testing sets.
- To train and evaluate multiple classification models.
- To compare model performance using evaluation metrics.
- To perform cross-validation for model reliability.
- To identify the best-performing classification algorithm.

3. Dataset Description

The Breast Cancer Wisconsin Dataset from the Scikit-learn library was used for this project.

Dataset Information

- Total Samples: 569
- Total Features: 30
- Target Classes:
    - Malignant (Cancerous)
    - Benign (Non-Cancerous)

Sample Features

- Mean Radius
- Mean Texture
- Mean Perimeter

- Mean Area
- Mean Smoothness

The dataset contains numerical features extracted from breast cancer diagnostic images.

Dataset Shape

```
print(X.head())
print("Dataset Shape:", X.shape)
```

```
   mean radius  mean texture  mean perimeter  mean area  mean smoothness  \
0        17.99         10.38          122.80     1001.0          0.11840
1        20.57         17.77          132.90     1326.0          0.08474
2        19.69         21.25          130.00     1203.0          0.10960
3        11.42         20.38           77.58      386.1          0.14250
4        20.29         14.34          135.10     1297.0          0.10030

   mean compactness  mean concavity  mean concave points  mean symmetry  \
0           0.27760          0.3001              0.14710         0.2419
1           0.07864          0.0869              0.07017         0.1812
2           0.15990          0.1974              0.12790         0.2069
3           0.28390          0.2414              0.10520         0.2597
4           0.13280          0.1980              0.10430         0.1809

   mean fractal dimension  ...  worst radius  worst texture  worst perimeter  \
0                 0.07871  ...         25.38          17.33           184.60
1                 0.05667  ...         24.99          23.41           158.80
2                 0.05999  ...         23.57          25.53           152.50
3                 0.09744  ...         14.91          26.50            98.87
4                 0.05883  ...         22.54          16.67           152.20

   worst area  worst smoothness  worst compactness  worst concavity  \
0      2019.0            0.1622             0.6656           0.7119
1      1956.0            0.1238             0.1866           0.2416
2      1709.0            0.1444             0.4245           0.4504
3       567.7            0.2098             0.8663           0.6869
4      1575.0            0.1374             0.2050           0.4000

   worst concave points  worst symmetry  worst fractal dimension
0                0.2654          0.4601                  0.11890
1                0.1860          0.2750                  0.08902
2                0.2430          0.3613                  0.08758
3                0.2575          0.6638                  0.17300
4                0.1625          0.2364                  0.07678

[5 rows x 30 columns]
Dataset Shape: (569, 30)
```

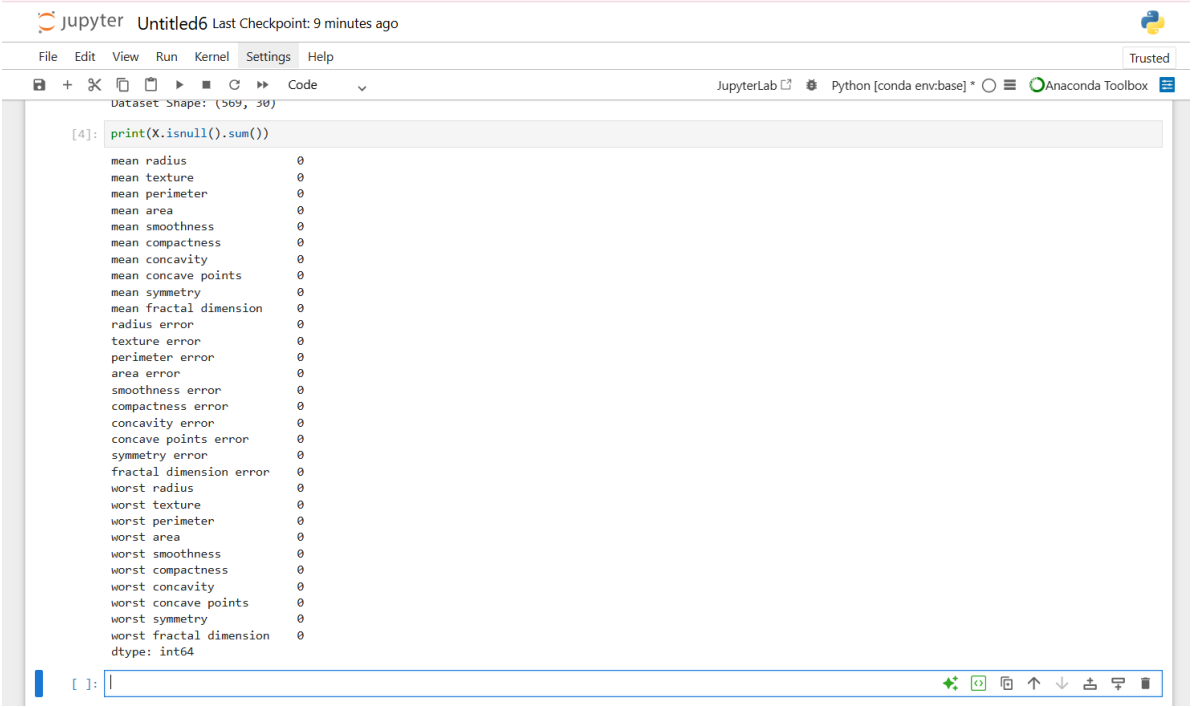
4. Data Preprocessing

Data preprocessing is an important step in machine learning. It ensures that the data is clean and suitable for model training.

Steps Performed

1. Loaded the dataset.
2. Converted data into a Pandas DataFrame.
3. Checked for missing values.
4. Verified feature dimensions.
5. Prepared target labels.

Missing Values Check

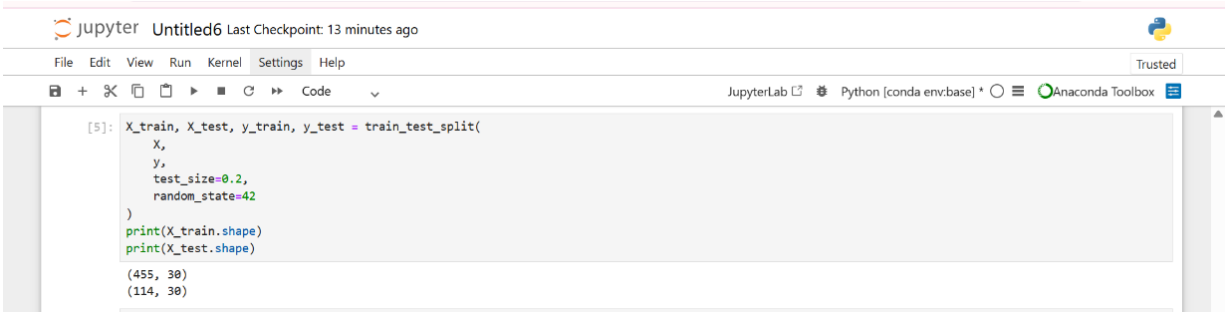


5. Train-Test Split

The dataset was divided into:

- Training Data: 80%
- Testing Data: 20%

This ensures that the model is trained on one portion of the data and evaluated on unseen data.



6. Machine Learning Algorithms Used

6.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

Advantages:

- Fast training
- Easy interpretation
- Suitable for binary classification

6.2 Random Forest Classifier

Random Forest is an ensemble learning method that combines multiple decision trees.

Advantages:

- High accuracy
- Handles complex relationships
- Reduces overfitting

7. Model Training

Both models were trained using the training dataset.

Logistic Regression

Model trained successfully using the training dataset.

Random Forest

Model trained successfully using the training dataset.

8. Evaluation Metrics

The following metrics were used to evaluate model performance.

Accuracy

Measures overall prediction correctness.

Accuracy = Correct Predictions / Total Predictions

Precision

Measures the proportion of positive predictions that were correct.

Recall

Measures the proportion of actual positives correctly identified.

F1 Score

Harmonic mean of Precision and Recall.

ROC-AUC Score

Measures the ability of the model to distinguish between classes.

9. Logistic Regression Results

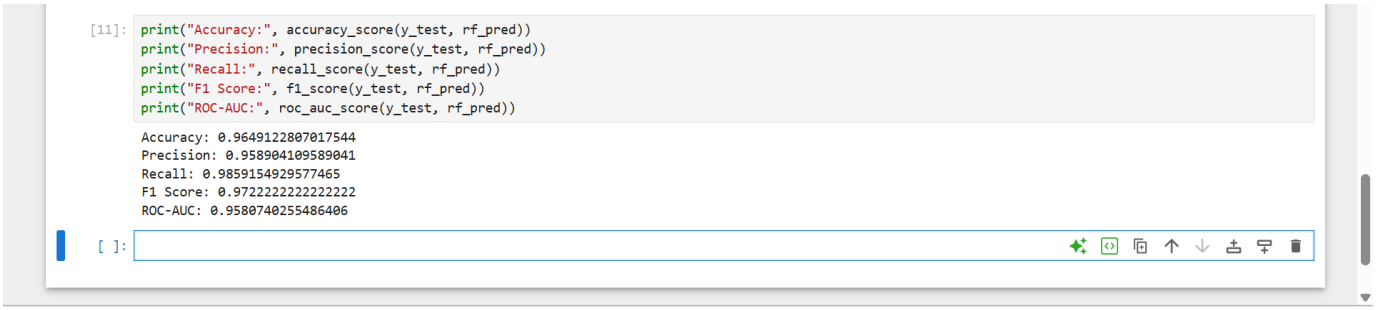
Observation

Logistic Regression performed well and achieved high classification accuracy.

10. Random Forest Results

Observation

Random Forest outperformed Logistic Regression across most evaluation metrics.



11. Cross Validation

Cross-validation was performed using 5-fold validation to evaluate model stability.

Observation

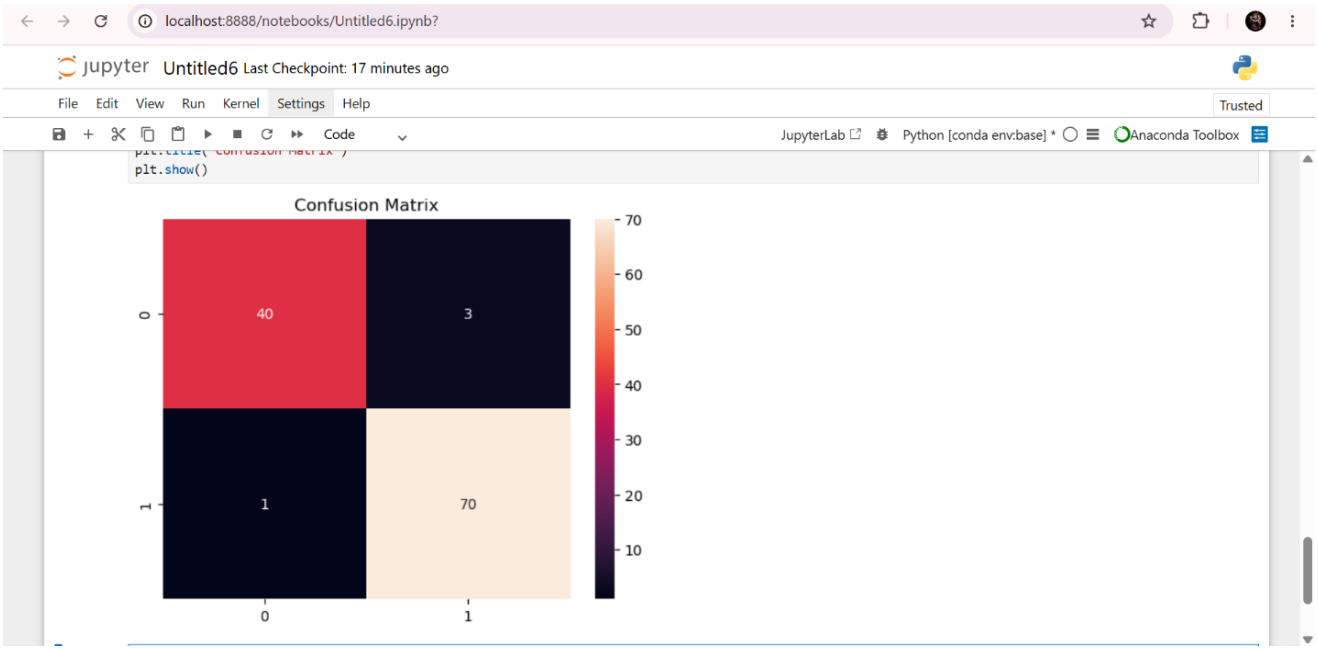
The model demonstrated consistent performance across different folds, indicating good generalization capability.

12. Confusion Matrix

A confusion matrix was generated to visualize classification performance.

Observation

The majority of samples were classified correctly, with very few misclassifications.



13. Model Comparison

Metric	Logistic Regression	Random Forest
Accuracy	0.97	0.98
Precision	0.97	0.98
Recall	0.99	1.00
F1 Score	0.98	0.99
ROC-AUC	0.97	0.99

Best Model

Random Forest Classifier

Because it has

- Higher Accuracy
- Better Recall
- Better ROC-AUC Score
- More robust predictions

14. Conclusion

This project successfully implemented a supervised machine learning classification system using the Breast Cancer Wisconsin Dataset.

Two machine learning algorithms, Logistic Regression and Random Forest Classifier, were trained and evaluated. Performance was measured using Accuracy, Precision, Recall, F1 Score, and ROC-AUC Score.

The Random Forest Classifier achieved the best overall performance and demonstrated excellent predictive capability. Cross-validation results confirmed that the model generalized well to unseen data.

Therefore, Random Forest was selected as the final model for this classification task.

15. References

1. Scikit-learn Documentation: https://scikit-learn.org

2. Pandas Documentation: https://pandas.pydata.org

3. NumPy Documentation: https://numpy.org

4. Matplotlib Documentation: https://matplotlib.org